Project: Worked examples
Title: Beam and 2D frame analysis
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Linear elastic analysis of a plane frame or continuous beam by the direct stiffness method: reactions, member end forces, axial force, shear, bending moment and deflection along every member with their extremes and positions, for each load combination and as an envelope. Global x to the right, y up; moments and couples anticlockwise positive; sagging moments positive along a member from node i to node j.

Three freedoms per node; Euler–Bernoulli member stiffness EA/L and EI/L³ terms; end releases by static condensation; supports by removing freedoms or adding spring stiffness; equivalent nodal loads from exact fixed-end forces; K u = F solved by banded LDL[1] after a reverse Cuthill–McKee ordering (a non-positive pivot is reported as a mechanism). Internal forces and deflections from the member end actions and Macaulay functions of the span loads; extremes at stations plus the zero-shear and zero-slope points found by bisection.

Ref: Hibbeler, Structural Analysis Ch. 16 — Plane frame analysis using the stiffness method
Ref: Roark's Formulas for Stress and Strain Ch. 8, Table 8.1 — Shear, moment, slope and deflection formulas for elastic straight beams
Ref: BS EN 1990 6.4.3 — Combinations of actions (user-entered factors)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
E	Default modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 3.2.6: E = 210 GPa	1 … 500 GPa
nodes	Nodes	3 row(s)		drawing: DRG-FL-201 rev A	any
nodes[1]		id = 1; x = 0 m; y = 0 m; support = pin			
nodes[2]		id = 2; x = 6 m; y = 0 m; support = roller (vertical)			
nodes[3]		id = 3; x = 12 m; y = 0 m; support = roller (vertical)			
elements	Elements	2 row(s)		reference: UB 305×165×40 (Blue Book)	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
elements[1]		id = 1; i = 1; j = 2; A = 51.3 cm^2; I = 8503 cm^4; rel_i = no; rel_j = no			
elements[2]		id = 2; i = 2; j = 3; A = 51.3 cm^2; I = 8503 cm^4; rel_i = no; rel_j = no			
loads	Loads	4 row(s)		spec: Load schedule LS-01: G = 8 kN/m, Q = 12 kN/m	any
loads[1]		case = 1; kind = udl; target = 1; a = 0 m; b = 6 m; w1 = -8 kN/m; w2 = 0 kN/m; F_x = 0 kN; F_y = 0 kN; M = 0 kN*m			
loads[2]		case = 1; kind = udl; target = 2; a = 0 m; b = 6 m; w1 = -8 kN/m; w2 = 0 kN/m; F_x = 0 kN; F_y = 0 kN; M = 0 kN*m			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
loads[3]		case = 2; kind = udl; target = 1; a = 0 m; b = 6 m; w1 = -12 kN/m; w2 = 0 kN/m; F_x = 0 kN; F_y = 0 kN; M = 0 kN*m			
loads[4]		case = 2; kind = udl; target = 2; a = 0 m; b = 6 m; w1 = -12 kN/m; w2 = 0 kN/m; F_x = 0 kN; F_y = 0 kN; M = 0 kN*m			
combinations	Combinations (blank = each case alone)	4 row(s)		spec: BS EN 1990 6.4.3.2 eq. 6.10 (γ_G 1.35, γ_Q 1.5); SLS characteristic	any
combinations[1]		name = ULS; case = 1; factor = 1.35			
combinations[2]		name = ULS; case = 2; factor = 1.5			
combinations[3]		name = SLS; case = 1; factor = 1			
combinations[4]		name = SLS; case = 2; factor = 1			
report	Combination to report in detail (blank = first)	ULS		assumption: ULS reported in detail; SLS in the envelope.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
n_div	Stations per member	24		assumption: 24 stations per span.	4 … 100

3. Assumptions

A1. Linear elastic, small displacements, first order; no shear deformation, no P–Δ, no buckling. (Euler–Bernoulli direct stiffness.)
A2. Members are prismatic between nodes and connect rigidly unless a release is stated; a release transmits no moment at that end. (Static condensation of the released rotation.)
A3. Distributed loads act perpendicular to the member unless marked as gravity loads, which act in global −y per metre of member length. (Load table convention.)
A4. report (Combination to report in detail (blank = first)) is taken as ULS. (ULS reported in detail; SLS in the envelope.)
A5. n_div (Stations per member) is taken as 24. (24 stations per span.)

4. Working

```
[4] n_nodes = nodes
= 3
= 3
```
Nodes in the model
```
[5] n_elements = elements
= 2
= 2
```
Members in the model
```
[6] n_comb = combinations
= ULS, SLS
= 2
```
Load combinations analysed
```
[7] comb = reported combination
= ULS
= "ULS"
```
The combination detailed below
```
[8] R1_x = reaction at node 1, x
= □ member end forces - applied nodal loads
= 0 kN
```
Horizontal reaction at node 1
```
[9] R1_y = reaction at node 1, y
= □ member end forces - applied nodal loads
= 64.8 kN
```
Vertical reaction at node 1
```
[10] R1_m = reaction at node 1, moment
= □ member end moments - applied nodal couple
= 0 kN·m
```
Moment reaction at node 1
```
[11] R2_x = reaction at node 2, x
= □ member end forces - applied nodal loads
= 0 kN
```
Horizontal reaction at node 2
```
[12] R2_y = reaction at node 2, y
= □ member end forces - applied nodal loads
= 216 kN
```
Vertical reaction at node 2

[13] R2_m = reaction at node 2, moment
= Σ member end moments − applied nodal couple
= 0 kN·m
Moment reaction at node 2
[14] R3_x = reaction at node 3, x
= Σ member end forces − applied nodal loads
= 0 kN
Horizontal reaction at node 3
[15] R3_y = reaction at node 3, y
= Σ member end forces − applied nodal loads
= 64.8 kN
Vertical reaction at node 3
[16] R3_m = reaction at node 3, moment
= Σ member end moments − applied nodal couple
= 0 kN·m
Moment reaction at node 3
[17] sum_R_y = Σ R_y
= all supports
= 345.6 kN
Total vertical reaction (equilibrium check against the applied load)
[18] sum_R_x = Σ R_x
= all supports
= 0 kN
Total horizontal reaction
[19] M_max_e1 = member 1: max sagging M
= at x = 2.25 m from node 1
= 72.9 kN·m
Member 1 maximum moment
[20] M_min_e1 = member 1: min (hogging) M
= at x = 6 m
= -129.6 kN·m
Member 1 minimum moment
[21] V_e1 = member 1: max |V|
= 108 kN
= 108 kN
Member 1 maximum shear
[22] N_e1 = member 1: max |N| (tension +)
= 0 kN
= 0 kN
Member 1 axial force of greatest magnitude
[23] d_e1 = member 1: max |deflection|
= at x = 2.5292 m
= 11.321 mm
Member 1 maximum transverse deflection
[24] M_max_e2 = member 2: max sagging M
= at x = 3.75 m from node 2
= 72.9 kN·m
Member 2 maximum moment
[25] M_min_e2 = member 2: min (hogging) M
= at x = 0 m
= -129.6 kN·m
Member 2 minimum moment

[26] V_e2 = member 2: max |V|
= 108 kN
= 108 kN
Member 2 maximum shear
[27] N_e2 = member 2: max |N| (tension +)
= 0 kN
= 0 kN
Member 2 axial force of greatest magnitude
[28] d_e2 = member 2: max |deflection|
= at x = 3.4708 m
= 11.321 mm
Member 2 maximum transverse deflection
[29] M_max = max sagging moment
= member 1 at x = 2.25 m
= 72.9 kN·m
Maximum moment in the reported combination
[30] x_Mmax = position of M_max
= from node i of member 1
= 2.25 m
Position of the maximum moment
[31] M_min = min (hogging) moment
= member 1 at x = 6 m
= -129.6 kN·m
Minimum moment in the reported combination
[32] V_max = max |V|
= member 1
= 108 kN
Maximum shear in the reported combination
[33] N_max = max tension
= member 1
= 0 kN
Maximum axial tension
[34] N_min = max compression (negative)
= member 1
= 0 kN
Maximum axial compression
[35] d_max = max |deflection|
= member 2 at x = 3.4708 m
= 11.321 mm
Maximum transverse deflection
[36] M_max_env = envelope max M
= ULS, member 1
= 72.9 kN·m
Maximum moment over all combinations
[37] M_min_env = envelope min M
= ULS, member 1
= -129.6 kN·m
Minimum moment over all combinations
[38] V_max_env = envelope max |V|
= ULS, member 1
= 108 kN
Maximum shear over all combinations

[39] d_max_env = envelope max |□|
= ULS, member 2
= 11.321 mm
Maximum deflection over all combinations

5. Results

SYMBOL	VALUE	DESCRIPTION
R1_x	0 kN	R_x node 1
R1_y	64.8 kN	R_y node 1
R1_m	0 kN·m	M node 1
R2_x	0 kN	R_x node 2
R2_y	216 kN	R_y node 2
R2_m	0 kN·m	M node 2
R3_x	0 kN	R_x node 3
R3_y	64.8 kN	R_y node 3
R3_m	0 kN·m	M node 3
sum_R_y	345.6 kN	Σ R_y
sum_R_x	0 kN	Σ R_x
M_max_e1	72.9 kN·m	M max, member 1
M_min_e1	-129.6 kN·m	M min, member 1
V_e1	108 kN	|V| max, member 1
N_e1	0 kN	N, member 1
d_e1	11.321 mm	δ max, member 1
M_max_e2	72.9 kN·m	M max, member 2
M_min_e2	-129.6 kN·m	M min, member 2
V_e2	108 kN	|V| max, member 2
N_e2	0 kN	N, member 2
d_e2	11.321 mm	δ max, member 2
M_max	72.9 kN·m	M max
x_Mmax	2.25 m	x of M max
M_min	-129.6 kN·m	M min
V_max	108 kN	|V| max
N_max	0 kN	N max (tension)
N_min	0 kN	N min (compression)
d_max	11.321 mm	δ max
M_max_env	72.9 kN·m	M max, envelope
M_min_env	-129.6 kN·m	M min, envelope
V_max_env	108 kN	|V| max, envelope

SYMBOL	VALUE	DESCRIPTION
d_max_env	11.321 mm	δ max, envelope

6. Checks (governing first)

No pass/fail checks in this calculation.

7. Applicability

No warnings.

8. Verification

Template struct.frame-2d@1.0.0 Content hash 43a986248a2918a3 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0